

Ch 5 Potential Quiz and Module 1 Exam Answers

A subset of these questions will appear on the Chapter 1-4 Quiz.

Quizzes are 30 minutes long and have 20 points.

A (probably different) subset of these questions will also appear on the Module 1 Exam, with questions from the Ch 1-4 and Ch 5 Quiz lists together making up at least half the points on the Module 1 Exam (i.e. at least 25 out of 50 points)

You should not expect Quiz or Exam questions to appear in the same order as in this document, nor should you expect the options for the answers to appear in the same order as in this document.

The remaining 50% of Module 1 Exam questions will be rephrasing or recombination of topics listed either in this document, in the Chapter 5 Potential Quiz questions, or the Ask Your Neighbor questions from class.

Questions are organized below into “pools” of similar questions; for each question it is shown how many points each of those questions would be worth should the question appear on the exam.

Answers are included for non-numeric questions in this document, but you should not count on being able to memorize an answer key for the exam, and you will not be allowed access to notes during the exam. You are strongly encouraged to form study groups to prepare for the exam, and you are encouraged to ask Gwen if you don't understand any concepts that are in this study sheet.

Pool 1: 1 Each worth 2 points

1. Math fill in the blank: Note that the values given for variables will not be the same values as shown here, and that exams will draw randomly from different combinations of numbers:

Answer both questions at the bottom about the following market:

$$U(C, l) = a \ln C + b \ln(l)$$

$$C = w(1 - t)(h - l)$$

Where $a = .5$; $b = .5$, $w(1-t) = 10$, and $h = 16$

A. Rounding to two decimal points, the representative utility-maximizing household will choose $C = [\text{Clevel}]$ (**enter number**).

B. In this example income effects are [relative] substitution effects (**enter >, <, or =**)

2. Math fill in the blank: Note that the values given for variables will not be the same values as shown here, and that exams will draw randomly from different combinations of numbers:

Answer both questions at the bottom about the following market:

$$U(C, l) = aC + b \ln(l)$$

$$C = w(1 - t)(h - l)$$

Where $a = .5$; $b = .5$, $w(1-t) = 10$, and $h = 16$

C. Rounding to two decimal points, the representative utility-maximizing household will choose $C = [\text{Clevel}]$ (**enter number**).

D. In this example income effects are [relative] substitution effects (**enter >, <, or =**)

Pool 2: Each worth 1 point.

1. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect an **increase in Government spending** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of leisure and to [*an increase/a decrease/no change/an ambiguous change*] in the leisure intercept for the Production Possibility Frontier (PPF).
2. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect an **increase in Government spending** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of employment and to [*an increase/a decrease/no change/an ambiguous change*] in welfare (utility).
3. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect an **increase in Government spending** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium real wages and to [*an increase/a decrease/no change/an ambiguous change*] in GDP.
4. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect a **decrease in Government spending** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of leisure and to [*an increase/a decrease/no change/an ambiguous change*] in the leisure intercept for the Production Possibility Frontier (PPF).

5. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect a **decrease in Government spending** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of employment and to [*an increase/a decrease/no change/an ambiguous change*] in welfare (utility).
6. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect a **decrease in Government spending** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium real wages and to [*an increase/a decrease/no change/an ambiguous change*] in GDP.

Pool 3: Each worth 1 point

1. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect an **increase in TFP (z)** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of leisure and to [*an increase/a decrease/no change/an ambiguous change*] in the leisure intercept for the Production Possibility Frontier (PPF).
2. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect an **increase in TFP (z)** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of employment and to [*an increase/a decrease/no change/an ambiguous change*] in welfare (utility).
3. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect an **increase in TFP (z)** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium real wages and to [*an increase/a decrease/no change/an ambiguous change*] in GDP.
4. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect a **decrease in TFP (z)** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of leisure and to [*an increase/a decrease/no change/an ambiguous change*] in the leisure intercept for the Production Possibility Frontier (PPF).

5. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect an **decrease in TFP (z)** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of employment and to [*an increase/a decrease/no change/an ambiguous change*] in welfare (utility).
6. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect an **decrease in TFP (z)** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium real wages and to [*an increase/a decrease/no change/an ambiguous change*] in GDP.

Pool 4: Each worth 1 point

1. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect an shift in preferences **toward Consumption** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of leisure and to [*an increase/a decrease/no change/an ambiguous change*] in the leisure intercept for the Production Possibility Frontier (PPF).
2. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect a shift in preference **toward Consumption** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of employment and to [*an increase/a decrease/no change/an ambiguous change*] in welfare (utility).
3. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect a shift in preferences **toward Consumption** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium real wages and to [*an increase/a decrease/no change/an ambiguous change*] in GDP.
4. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect an shift in preferences **toward leisure** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of leisure and to [*an increase/a decrease/no change/an ambiguous change*] in the leisure intercept for the Production Possibility Frontier (PPF).

5. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect a shift in preference **toward leisure** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium levels of employment and to [*an increase/a decrease/no change/an ambiguous change*] in welfare (utility).
6. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect a shift in preferences **toward leisure** to lead to [*an increase/a decrease/no change/an ambiguous change*] in equilibrium real wages and to [*an increase/a decrease/no change/an ambiguous change*] in GDP.

Pool 5: Each worth 1 point

1. Assume that **TFP (z) is a positive (increasing) function of Government spending (G)**. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect **an increase in G** to lead to *[an increase/a decrease/no change/an ambiguous change]* in equilibrium levels of leisure **compared to the case where G is unproductive**, and to *[an increase/a decrease/no change/an ambiguous change]* in the leisure intercept for the Production Possibility Frontier (PPF) **compared to the case where G is unproductive**.
2. Assume that **TFP (z) is a positive (increasing) function of Government spending (G)**. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect **an increase in G** to lead to *[an increase/a decrease/no change/an ambiguous change]* in equilibrium levels of employment **compared to the case where G is unproductive**, and to *[an increase/a decrease/no change/an ambiguous change]* in welfare (utility) **compared to the case where G is unproductive**.
3. Assume that **TFP (z) is a positive (increasing) function of Government spending (G)**. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect **an increase in G** to lead to *[an increase/a decrease/no change/an ambiguous change]* in equilibrium real wages **compared to the case where G is unproductive**, and to *[an increase/a decrease/no change/an ambiguous change]* in GDP **compared to the case where G is unproductive**.
4. Assume that **TFP (z) is a positive (increasing) function of Government spending (G)**. *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect **a decrease in G** to lead to *[an increase/a decrease/no change/an ambiguous change]* in equilibrium levels of leisure **compared to the case where G is unproductive**, and to *[an increase/a decrease/no change/an ambiguous change]* in the leisure intercept for the Production Possibility Frontier (PPF) **compared to the case where G is unproductive**.

5. Assume that $TFP(z)$ is a positive (increasing) function of Government spending (G). *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect a **decrease in G** to lead to *[an increase/a decrease/no change/an ambiguous change]* in equilibrium levels of employment **compared to the case where G is unproductive**, and to *[an increase/a decrease/no change/an ambiguous change]* in welfare (utility) **compared to the case where G is unproductive**.

6. Assume that $TFP(z)$ is a positive (increasing) function of Government spending (G). *Ceteris paribus*, and using the Social Planner's model from Chapter 5, we expect a **decrease in G** to lead to *[an increase/a decrease/no change/an ambiguous change]* in equilibrium real wages **compared to the case where G is unproductive**, and to *[an increase/a decrease/no change/an ambiguous change]* in GDP **compared to the case where G is unproductive**.

Pool 6: Each worth 1 point

1. Assuming as in class that Consumption is on the vertical axis, the presence of a [**proportional income tax/lump sum tax**] will result in MRS out of consumption into leisure [**less than/greater than**] MPN
2. In the one-period general equilibrium model from class, changes in government spending predict that real wages will move [**pro-cyclically/counter-cyclically/ambiguously**] as GDP moves, whereas in the US data real wages move [**pro-cyclically/counter-cyclically/ambiguously**] as GDP moves.
3. In the one-period general equilibrium model from class, changes in TFP (z) predict that real wages will move [**pro-cyclically/counter-cyclically/ambiguously**] as GDP moves, whereas in the US data real wages move [**pro-cyclically/counter-cyclically/ambiguously**] as GDP moves.
4. According to the one-period general equilibrium model from class, if the government replaces a lump sum tax with a proportional labor income tax, then employment will [**increase/decrease/either increase or decrease**] whereas GDP will [**increase/decrease/either increase or decrease**].
5. Assuming that substitution effects dominate income effects in the one-period general equilibrium model from class, an **increase** in the proportional income tax rate will [**always/sometimes/never**] lead to a **decrease** in tax revenue and will [**always/sometimes/never**] lead to a **decrease** in GDP.
6. Assume that there is a **positive shock to TFP** in the one-period general equilibrium model from class. This model will do a better job of predicting short-run (business cycle) movements in hours worked if substitution effects [**dominate/are dominated by/are exactly offset by**] income effects and will do a better job predicting long-run (more than a few years) movements in hours worked if substitution effects [**dominate/are dominated by/are exactly offset by**] income effects

Pool 7: Each worth 1 point:

1. In our general equilibrium model, *[Employment/Leisure/Consumption/**Government Spending**]* is thought to be exogenous.

2. If two data series are negatively correlated it means that:

- A. There is no relationship between the data
- B. If one data series is increasing, the other is also increasing
- C. If one data series is decreasing, the other is increasing
- D. If one data series is decreasing, the other is also decreasing

Answer: C

3. In the business cycle data presented in class, GDP is negatively correlated with *[Consumption/Real Wage/Total Factor Productivity/**Leisure**]*

4. In the general equilibrium model, if preferences change towards desiring more leisure, the PPF curve will *[get steeper/get flatter/**not change**]*

5. In the competitive equilibrium of our one-period model, which of the following is not always true even if the equilibrium is Pareto Optimal?

- A. $MPN = w$
- B. $MRT = w$
- C. $MU_l / MU_c = MRT$
- D. $MPN = z$

Answer: D

6. Which of the following is a proper formulation of the household budget constraint in our model when $\pi=0$ and $T=0$?

- A. $h = C + l$
- B. $h = l + l/w * c$
- C. $wh = l + c$
- D. $h = wl + c$

Answer: B

7. When the government is using a proportional tax on labor income, MRS should be equal to $[MPN/w/MRT/w(1-t)]$

8. Based on our one-period model from class, an increase in lump-sum taxes (T) causes

- A. Negative income effects.
- B. Positive income effects.
- C. Substitution effects from leisure to consumption
- D. No effect on the model.

Answer: A

9. Our general equilibrium model is most consistent with the correlations in the data when we have shocks to:

- A. G/T
- B. Preferences
- C. Prices
- D. Z

Answer: D

10. Which of the following has a substitution effect of zero in the one-period partial-equilibrium model of chapter 4 (i.e. pure income effect)?

- A. Wage changes
- B. Firms become more profitable
- C. Firms become more productive
- D. Preferences shift

Answer: B

11. In our one-period consumer model, if $MU_l/MU_c > w$ a consumer should

- A. Work more
- B. Work less
- C. Work the same amount
- D. Try to increase their wage

Answer: B

12. In which one of the following utility functions does leisure not depend on the level of a payroll tax (proportional income tax)?

- A. $\frac{1}{2}c + \frac{1}{2}l$
- B. $\frac{1}{2}\ln(C) + \frac{1}{2}\ln(l)$
- C. $\frac{3}{4}c + \frac{1}{4}l$
- D. $\frac{3}{4}\ln(C) + \frac{1}{4}l$

Answer: B

13. Consider the model from class. According to the model, shocks to preferences cannot be a source for business cycle fluctuations because the model counter-factually predicts

- A. Countercyclical consumption.
- B. Countercyclical wages.
- C. Pro-cyclical labor hours.
- D. Pro-cyclical technology

Answer: B